

Linear Algebra — Exercises

Systems of linear equations

Exercise 1. Find a basis and the dimension of the solution space over the field $\mathbb{R}$ of each of the following systems of linear equations:

$$\begin{cases} x + 2y - 2z + 2s - t = 0 \\ x + 2y - z + 3s - 2t = 0 \\ 2x + 4y - 7z + s + t = 0 \end{cases} \quad \begin{cases} x + 2y - z + 3s - 4t = 0 \\ 2x + 4y - 2z - s + 5t = 0 \\ 2x + 4y - 2z + 4s - 2t = 0 \end{cases}$$

Exercise 2. Discuss the solvability of the following systems of equations over the field $\mathbb{R}$ in terms of the parameters a and b :

$$\begin{cases} ax + by + az + t = 1 \\ ax + y + az + t = 1 \\ x + y + az + at = 1 \end{cases} \quad \begin{cases} x + 2y + 3z + 4t = a \\ 2x + 3y + 4z + 5t = b \\ 3x + 4y + 5z + 6t = a \\ 4x + 5y + 6z + 7t = b \end{cases} \quad \begin{cases} x + ay + az = 1 \\ ax + ay + z = 1 \\ ax + y + az = 1 \\ ax + ay + az = 1 \end{cases}$$

Exercise 3. Find general solutions of the following systems of equations over the field $\mathbb{R}$:

$$(a) \begin{cases} 2x + 2y + 3z + 4t = 2 \\ 2x + 3y + 4z + 6t = 0 \\ 3x + 4y + 5z + 6t = 1 \\ x + y + z + t = 1 \end{cases}$$

$$(b) \begin{cases} x + y + z - t + u = 1 \\ x - y + z + t + u = 1 \\ -x + y - z + t - u = 0 \end{cases}$$

$$(c) \begin{cases} x + y + z + t + u = 0 \\ x - 2y + 3z - 4t + 5u = 0 \end{cases}$$

Exercise 4. Find all solutions in $\mathbb{Z}_7$ of the following system of equations:

$$\begin{cases} 3x_1 + 4x_2 + 4x_3 + 6x_4 = 1 \\ 4x_1 + 5x_2 + 4x_3 + x_4 = 0 \\ 5x_1 + 6x_2 + 4x_3 + 3x_4 = 6 \end{cases} \quad \text{over the field } \mathbb{Z}_7.$$

Exercise 5. Calculate the following determinants:

$$(a) \det \begin{pmatrix} 1 & 2 & -2 & 1 \\ -3 & 1 & 3 & 2 \\ -1 & 2 & 4 & 2 \\ 5 & -1 & 2 & 2 \end{pmatrix}, \text{ where the matrix belongs to } \mathbb{R}_4^4$$

$$(b) \det \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 1 \\ 3 & 2 & 3 & 2 \\ 4 & 3 & 4 & 3 \end{pmatrix}, \text{ where the matrix belongs to } (\mathbb{Z}_5)_4^4$$

$$(c) \det(A), \text{ where } A \in \mathbb{K}_5^5 \text{ has the entries } a_{ij} = i - j.$$

Exercise 6. The numbers 24026, 40262, 2624, 26240 and 62402 are divisible by 41. Prove that the following determinant is also divisible by 41:

$$\det \begin{pmatrix} 2 & 4 & 0 & 2 & 6 \\ 4 & 0 & 2 & 6 & 2 \\ 0 & 2 & 6 & 2 & 4 \\ 2 & 6 & 2 & 4 & 0 \\ 6 & 2 & 4 & 0 & 2 \end{pmatrix}.$$

Exercise 7. Let $y \in \mathbb{R}$, $n \in \mathbb{N}$ and let $A \in \mathbb{R}_n^n$ be defined in the following way: $a_{ij} = y$ for $j \neq n - i + 1$ and $a_{ij} = 0$ for $j = n - i + 1$. Find the determinant of A for $n = 3$, and then find the general formula for the determinant of A for an arbitrary $n \in \mathbb{N}$. Justify your answer.

Exercise 8. Show that for all matrices A and B of size 2×2 , $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Exercise 9. Show, for invertible matrices A and B , that $(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$.

Exercise 10. Calculate the inverse (if it exists) of each of the following matrices:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in (\mathbb{Z}_3)_4^4, \quad B = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 1 & 3 & 4 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 3 & 4 & 2 \end{pmatrix} \in \mathbb{R}_4^4, \quad C = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 1 & 3 & 4 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 3 & 4 & 2 \end{pmatrix} \in (\mathbb{Z}_5)_4^4.$$